INFORMATION TECHNOLOGIES
DEVELOPMENT OF
CROSS-PLATFORM SOFTWARE
AREA

INTEGRAL CALCULUS WORKSHOP

"HW4: INTEGRATION BY PARTIAL FRACTIONS"

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HW4: Integration by partial fractions

1

$$\int \frac{12}{(x^2-9)}\, dx$$

$$\int \frac{12}{x^2-9}\, dx = \int \frac{12}{(x-3)(x+3)}\, dx = \frac{A}{x-3} + \frac{B}{x+3}$$

$$12 = A(x+3) + B(x-3)$$

$$12 = Ax + 3A + Bx - 3B = \underline{(A+B)}x + \underline{(3A-3B)} =$$

$$A+B = 0 = \underline{B = -A}$$
$$3A - 3B = 12$$

$$3A - 3(-A) = 12 = 3A + 3A = 12$$

$$6A = 12 \qquad B = -A = B = -2$$
$$A = 2$$

$$\int \left(\frac{2}{x-3} - \frac{2}{x+3}\right) dx = 2\int \frac{1}{x-3}\, dx - 2\int \frac{1}{x+3}\, dx$$

$$2\ln|x-3| - 2\ln|x+3| + C$$

$$2\ln\left|\frac{x-3}{x+3}\right| + C = \underline{\ln\left|\left(\frac{x-3}{x+3}\right)^2\right| + C}$$

2.

$$\int \frac{4(x-4)}{(x^2-2x-3)}\,dx$$

$$\int \frac{4(x-4)}{(x-3)(x+1)}\,dx = \frac{A}{x-3} + \frac{B}{x+1}$$

$$4(x-4) = A(x+1) + B(x-3)$$

$$4x - 16 = Ax + A + Bx - 3B = (A+B)x + (A-3B)$$

$$A+B = 4 \quad = A = 4 - B \quad = A = 4 - 5 = -1$$

$$A - 3B = -16$$

$$(4-B) - 3B = -16$$

$$4 - 4B = -16$$

$$-4B = -4 - 16$$

$$-4B = -20$$

$$B = 5$$

$$\int \frac{-1}{x-3} + \frac{5}{x+1}\,dx = -\int \frac{1}{x-3}\,dx + 5\int \frac{1}{x+1}\,dx$$

$$-\ln|x-3| + 5\ln|x+1| + C$$

$$\ln\left|\frac{(x+1)^5}{x-3}\right| + C$$

3.

$$\int \frac{3(\overset{x}{2x^2 - 8x - 1})}{(x+4)(x+1)(2x-1)}\,dx$$

$$\int \frac{6x^2 - 24x - 3}{(x+4)(x+1)(2x-1)}\,dx = \frac{A}{x+4} + \frac{B}{x+1} + \frac{C}{2x-1}$$

$$6x^2 - 24x - 3 = A(x+1)(2x-1) + B(x+4)(2x-1)$$
$$+ C(x+4)(x+1)$$

$$A(x+1)(2x-1) = A[2x^2 + x - 1]$$

$$B(x+4)(2x-1) = B[2x^2 + 7x - 4]$$

$$C(x+4)(x+1) = C[x^2 + 5x + 4]$$

$$(2A + 2B + C)x^2 + (A + 7B + 5C)x + (-A - 4B + 4C)$$

$6x^2 = 2A + 2B + C = 6$ $\qquad$ $A = 3$

$-24x = A + 7B + 5C = -24$ $\qquad$ $B = -6$

$-3 = -A - 4B + 4C = -3$ $\qquad$ $C = 9$

$$\int \left(\frac{3}{x+4} - \frac{6}{x+1} + \frac{9}{2x-1}\right) dx$$

$$3\ln|x+4| - 6\ln|x+1| + \frac{9}{2}\ln|2x-1| + C$$

$$\ln\left(\frac{(x+4)^3 (2x-1)^{9/2}}{(x+1)^6}\right) + C = \underline{\ln\left(\frac{(x+4)^7}{(x+1)^3(2x-1)^2}\right) + C}$$

4

$$\int \frac{x^2+9x+8}{x^2+x-6}\,dx$$

$$\int \frac{x^2+9x+8}{(x+3)(x-2)}\,dx = 1+\frac{8x+14}{(x+3)(x-2)}$$

$$\int 1+\frac{8x+14}{(x+3)(x-2)}\,dx = \int 1\,dx + \int \frac{8x+14}{(x+3)(x-2)}\,dx$$

$$\frac{8x+14}{(x+3)(x-2)} = \frac{A}{x+3} + \frac{B}{x-2}$$

$$8x+14 = A(x-2)+B(x+3)$$

$$8x+14 = Ax-2A+Bx+3B = (A+B)x+(-2A+3B)$$

$$A+B=8 = A=8-B$$
$$-2A+3B=14$$

$$-2(8-B)+3B=14 = -16+2B+3B=14$$
$$5B=30 = B=6$$
$$A=8-6=2$$

$$\int\left(1+\frac{2}{x+3}+\frac{6}{x-2}\right)dx = x+2\ln|x+3|+6\ln|x-2|+C$$

$$x+\ln\left((x+3)^2(x-2)^6\right)+C$$

5. $\int \frac{4x-3}{(x+1)^2}\,dx \qquad u = x+1$

$$\int \frac{4x-3}{(x+1)^2}\,dx = \int \frac{4(u-1)-3}{u^2}\,du = \int \frac{4u-4-3}{u^2}\,du$$

$$\int \frac{4u-7}{u^2}\,du = \int\left(\frac{4u}{u^2} - \frac{7}{u^2}\right)du = \int\left(\frac{4}{u} - 7u^{-2}\right)du$$

$$\int\left(\frac{4}{u} - 7u^{-2}\right)du = 4\ln|u| + 7u^{-1} + C$$

$$\underline{4\ln|x+1| + \frac{7}{x+1} + C}$$

6.

$$\int \frac{5x^2-30x+44}{(x-2)^3}\,dx$$

$$5x^2-30x+44 = 5(x+2)^2 - 30(x+2) + 44$$

$(x+2)^2 = x^2 + 4x + 4$

$5(x+2)^2 = 5x^2 + 20x + 20$

$-30(x+2) = -30x - 60$

$$5x^2-30x+44 = 5x^2+20x+20-30x-60+44 = 5x^2-10x+4$$

$$\int \frac{5x^2-10x+4}{(x-2)^3}\,dx = \int\left(\frac{5}{x+2} - \frac{10}{(x+2)^2} + \frac{4}{(x-2)^3}\right)dx$$

$$\underline{5\ln|x-2| + \frac{10}{x+2} - \frac{2}{(x-2)^2} + C}$$

7. $\int_1^2 \frac{x^2+7x+3}{x^2(x+3)}\,dx$

$\frac{A}{x}+\frac{B}{x^2}+\frac{C}{x+3} = x^2+7x+3 = Ax(x+3)+B(x+3)Cx^2$

$Ax(x+3) = Ax^2+3Ax$
$B(x+3) = Bx+3B$
Cx^2

$x^2+7x+3 = (A+C)x^2+(3A+B)x+3B$

$A+C=1$ $\quad$ $B=1$
$3A+B=7$ $\quad$ $3A+1=7 = A=2$
$3B=3 = B=1$ $\quad$ $A+C=1 = 2+C=1 \quad C=-1$

$\int_1^2 \left(\frac{2}{x}+\frac{1}{x^2}-\frac{1}{x+3}\right)dx$

$\int_1^2 \frac{2}{x}\,dx + \int_1^2 \frac{1}{x^2}\,dx - \int_1^2 \frac{1}{x+3}\,dx$

$\int_1^2 \frac{2}{x}\,dx = 2\left|\ln x\right|_1^2 = 2(\ln 2-\ln 1) = 2\ln 2 = 2(0.6931) = 1.3862$

$\int_1^2 \frac{1}{x^2}\,dx = \int_1^2 x^{-2}\,dx = \left[-x^{-1}\right]_1^2 = -\frac{1}{2}+1 = \frac{1}{2}$

$\int_1^2 \frac{1}{x+3}\,dx = \left[\ln(x+3)\right]_1^2 = \ln(5)-\ln(4) = \ln\left(\frac{5}{4}\right)$

$\ln(1.25) = 0.2231$

$1.3862+0.5-0.2231 = \underline{1.6631}$

8. $\displaystyle\int \frac{x^2 - x - 13}{(x^2+7)(x-2)}\,dx$

$\dfrac{Ax+B}{x^2+7} + \dfrac{C}{x-2}$ $\qquad x^2 - x - 13 = (Ax+B)(x-2) + C(x^2+7)$

$(Ax+B)(x-2) = Ax^2 - 2Ax + Bx - 2B = Ax^2 + (B-2A)x - 2B$

$C(x^2+7) = (x^2+7)C$

$x^2 - x - 13 = (A+C)x^2 + (B-2A)x + (-2B+7C)$

$A + C = 1 \qquad\qquad A + C = 1 \qquad C = 1 - A$

$B - 2A = -1 \qquad\quad B - 2A = -1 \qquad B = 2A - 1$

$-2B + 7C = -13 \qquad -2B + 7C = -13$

$-2(2A-1) + 7(1-A) = -13 = -4A + 2 + 7 - 7A = -13$

$-11A + 9 = -13 \qquad -11A = -22 \qquad A = 2$

$A = 2$

$C = 1 - A = -1 \qquad \dfrac{x^2 - x - 13}{(x^2+7)(x-2)} = \dfrac{2x+3}{x^2+7} - \dfrac{1}{x-2}$

$B = 2A - 1 = 3$

$\displaystyle\int \left(\frac{2x+3}{x^2+7} - \frac{1}{x-2}\right) dx = \int \frac{2x}{x^2+7}\,dx + \int \frac{3}{x^2+7}\,dx$

$\displaystyle - \int \frac{1}{x-2}\,dx \qquad \int \frac{3}{x^2+(\sqrt{7})^2}\,dx = \frac{3}{\sqrt{7}} \tan^{-1}\left(\frac{x}{\sqrt{7}}\right)$

$\ln(x^2+7) + \dfrac{3}{\sqrt{7}} \tan^{-1}\left(\dfrac{x}{\sqrt{7}}\right) - \ln(x-2) + C$

9. $\int_5^6 \frac{6x-5}{(x-4)(x^2+3)} dx$

$$\frac{6x-5}{(x-4)(x^2+3)} = \frac{A}{x-4} + \frac{Bx+C}{x^2+3}$$

$$6x-5 = A(x^2+3)+(Bx+C)(x-4)$$

$$A(x^2+3) = Ax^2+3A$$
$$(Bx+C)(x-4) = Bx^2-4Bx+Cx-4C = Bx^2+(C-4B)x-4C$$

$$6x-5 = (A+B)x^2+(C-4B)x+(3A-4C)$$

$$B=-1 \quad A=1 \quad C=6-4(1)=2$$

$$C-4(-A)=6 \quad C+4A=6 \quad C=6-4A$$

$$3A-4(6-4A)=-5 \quad 3A-24+16A=-5 \quad 19A=19 \quad A=1$$

$$\frac{1}{x-4}+\frac{-x+2}{x^2+3} \quad \int_5^6\left(\frac{1}{x-4}+\frac{-x+2}{x^2+3}\right)dx$$

$$\int_5^6 \frac{1}{x-4}dx + \int_5^6 \frac{-x}{x^2+3}dx + \int_5^6 \frac{2}{x^2+3}dx$$

$$\ln|x-4|\Big|_5^6 = \ln(2)-\ln(1) = \ln(2)$$

$$-\frac{1}{2}\ln(x^2+3)\Big|_5^6 = -\frac{1}{2}[\ln(6^2+3)-\ln(5^2+3)] = -\frac{1}{2}(\ln(39)-\ln(28))$$
$$= -\frac{1}{2}\ln\left(\frac{39}{28}\right)$$

$$2\int_5^6 \frac{1}{x^2+(\sqrt{3})^2}dx = \frac{2}{\sqrt{3}}\left[\tan^{-1}\left(\frac{x}{\sqrt{3}}\right)\right]_5^6$$

$$\underline{0.5880}$$

$$\ln(2)-\frac{1}{2}\left(\frac{39}{28}\right)+\frac{2}{\sqrt{3}}\left[\tan^{-1}\left(\frac{6}{\sqrt{3}}\right)-\tan^{-1}\left(\frac{5}{\sqrt{3}}\right)\right]$$

10)

$$\int_1^2 \frac{4}{16-x^2}\,dx$$

$$\int \frac{4}{16-x^2} = \frac{4}{(4-x)(4+x)} = \frac{4}{4^2-x^2}$$

$$\int \frac{4}{16-x^2}\,dx = 4\int \frac{1}{16-x^2}\,dx = 4\cdot\frac{1}{8}\ln\left|\frac{4+x}{4-x}\right|$$

$$= \frac{1}{2}\ln\left|\frac{4+x}{4-x}\right| \qquad \left[\frac{1}{2}\ln\left|\frac{4+x}{4-x}\right|\right]_1^2$$

$$\frac{4+2}{4-2} = \frac{6}{2} = 3 \quad \ln(3)$$

$$\frac{4+1}{4-1} = \frac{5}{3} \quad \ln\left(\frac{5}{3}\right)$$

$$\frac{1}{2}[\ln(3) - \ln(5/3)] = \frac{1}{2}\ln\left(\frac{3}{5/3}\right) = \frac{1}{2}\ln\left(\frac{9}{5}\right)$$

$$\ln\left(\frac{9}{5}\right) = \ln(1.8) = 0.5878$$

$$\frac{1}{2}\cdot 0.5878 = \underline{0.2939}$$